Sahaj Bhadja

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Summary

Software Engineer pursuing an M.S. in Computer Science at USC with 2+ years building **TypeScript/React** interfaces, scalable **telemetry-style data pipelines**, and reliable **cloud-native backend services** with strong testing, documentation, and Agile delivery practices.

Education

University of Southern California **Los Angeles, CA**
Master of Science in Computer Science *August 2024 – May 2026*
GPA: 3.75/4 Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University **Gandhinagar, India**
B.Tech. in Computer Science and Engineering *November 2020 – May 2024*
GPA: 3.88/4 Coursework: OOP, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*

- Built **Python** workflow services and data interfaces for **500+ users**, reducing latency from **1.2s to 350ms** through query optimization while improving backend observability and cloud data-access reliability.
- Implemented fault-tolerant orchestration for **5K+ monthly jobs** with validation safeguards and rollback checks, improving **pipeline reliability**, data availability, and issue triage across distributed engineering workflows.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*

- Developed **React** and **Flask** features handling **10K+ daily requests**, translating requirements into user-friendly components and secure APIs while improving telemetry-style operational visibility and defect traceability.
- Improved release velocity with **CI/CD** controls and concurrent processing, reducing turnaround time by **60%** while preserving code quality through maintainable tests, automated checks, and documented fixes.

USC AI Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **Python** experimentation pipelines over **100K+ trajectory logs**, enabling rigorous behavior analysis and reproducible benchmark runs that strengthened large-scale data processing and engineering decision-making quality.
- Produced technical summaries and debugging diagnostics that converted findings into actionable tasks, improving **cross-functional collaboration** and reducing ambiguity in engineering handoffs, timelines, and issue-resolution planning.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **API** and **ETL** microservice workflows with OOP, AWS Glue, and S3, improving response times by **30%** while supporting scalable integrations and dependable data exchange across distributed systems.
- Maintained **20+** unit and integration suites with Jenkins, PyTest, and Mockito across **Linux/Windows** environments, boosting pipeline efficiency by **68%** and reducing technical debt by **20%** through automation.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **REST API** and **WebSocket** modules supporting eight workflows and **2M+ synchronized records**, improving service reliability and user-facing responsiveness for high-throughput, data-driven operations.
- Automated deployments and testing with **Docker** and **Jenkins**, reducing deployment time by **50%** and improving release consistency through script-driven validation and robust rollback mechanisms.

Skills

Languages: Python, JavaScript, TypeScript, Java, C++, SQL, Bash
Frontend: React, TypeScript, JavaScript, Responsive UI Development
Backend & Data: Flask, FastAPI, REST APIs, Microservices, ETL Pipelines, Data Management
Cloud & Infra: AWS (S3, Glue, ECS), Azure, Kubernetes, Docker, Linux/UNIX
Engineering Practices: Unit/Integration Testing, CI/CD, Git, Jenkins, GitHub Actions, Agile/Scrum

Projects

FinNexus | *Next.js, TypeScript, Kubernetes, MongoDB, Plaid*

- Built a scalable fintech platform supporting **50K+ users** and **1M+ transactions** with secure integrations, real-time updates, and **Kubernetes**-backed deployment workflows centered on reliability and operational performance.

Linguistix | *Python, AWS, Docker, CI/CD*

- Developed and deployed a containerized **RESTful API** on AWS CodePipeline and ECS/Fargate with automated **CI/CD** and validation checks, achieving **87% accuracy** and production-grade reliability at scale.

Chat2SQL | *Java, TypeScript, SQL, Docker, Maven*

- Built an AI-assisted SQL workspace supporting **10+ databases**, improving query authoring and schema exploration through structured data workflows, repeatable validations, and robust multi-environment troubleshooting capabilities.